

The background features a dark, textured surface resembling asphalt. On the left side, there are several stylized road arrows: a straight vertical arrow pointing up, and three curved arrows pointing in various directions (up-left, down-left, and up-right). The top and bottom edges of the image are framed by yellow and black diagonal hazard stripes.

Team 6

Bicycle Safety

INTRODUCTION

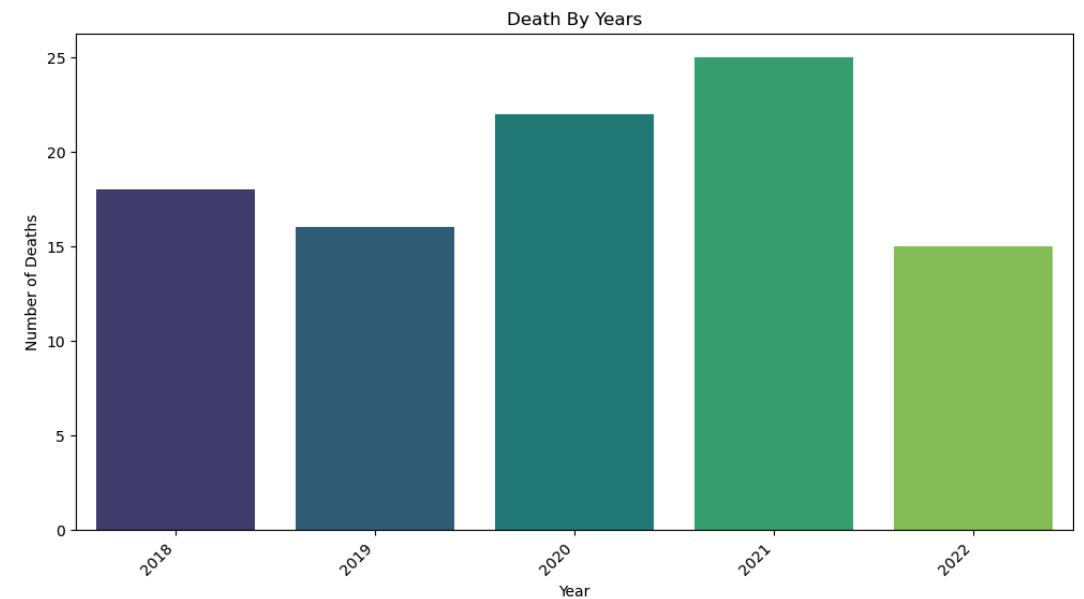
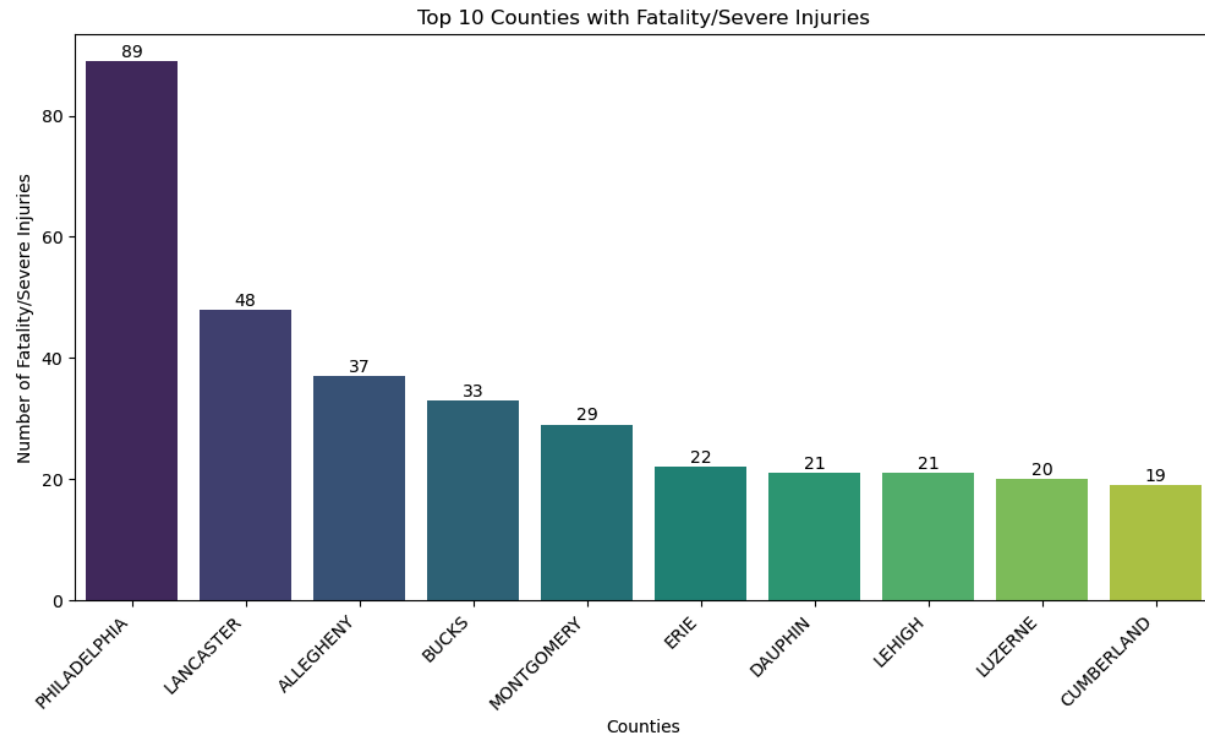
Business Question

Investigating the critical elements elevating
the risk for bicyclists



OVERVIEW

- Key findings of a report on bicycle safety in Pennsylvania. The report analyzed data from 2018 to 2022 to identify the factors that contribute to bicycle fatalities and severe injuries.



KEY INSIGHTS

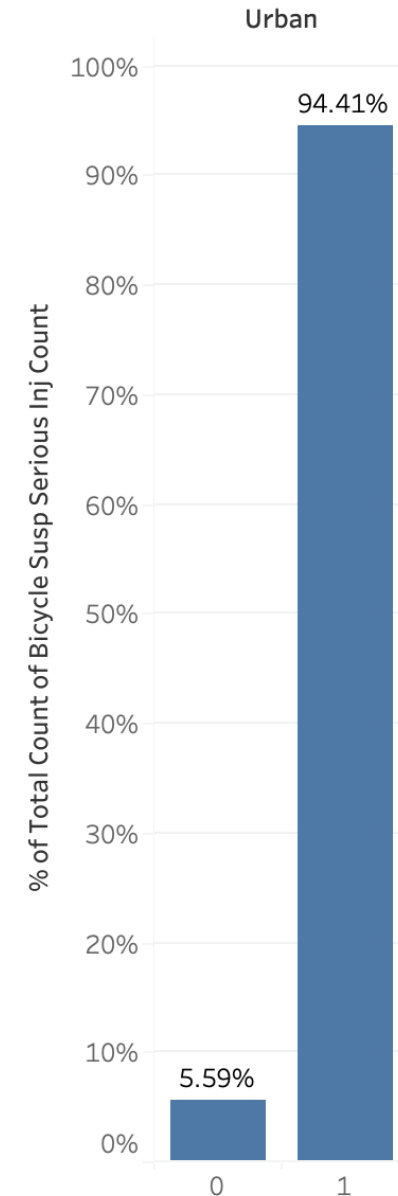
- Urban vs. Rural Dynamics
- Intersection Dynamics
- Stop Signs as Lifesavers
- Signalized Intersections
- Additional Insights – Time Series



URBAN VS. RURAL DYNAMICS

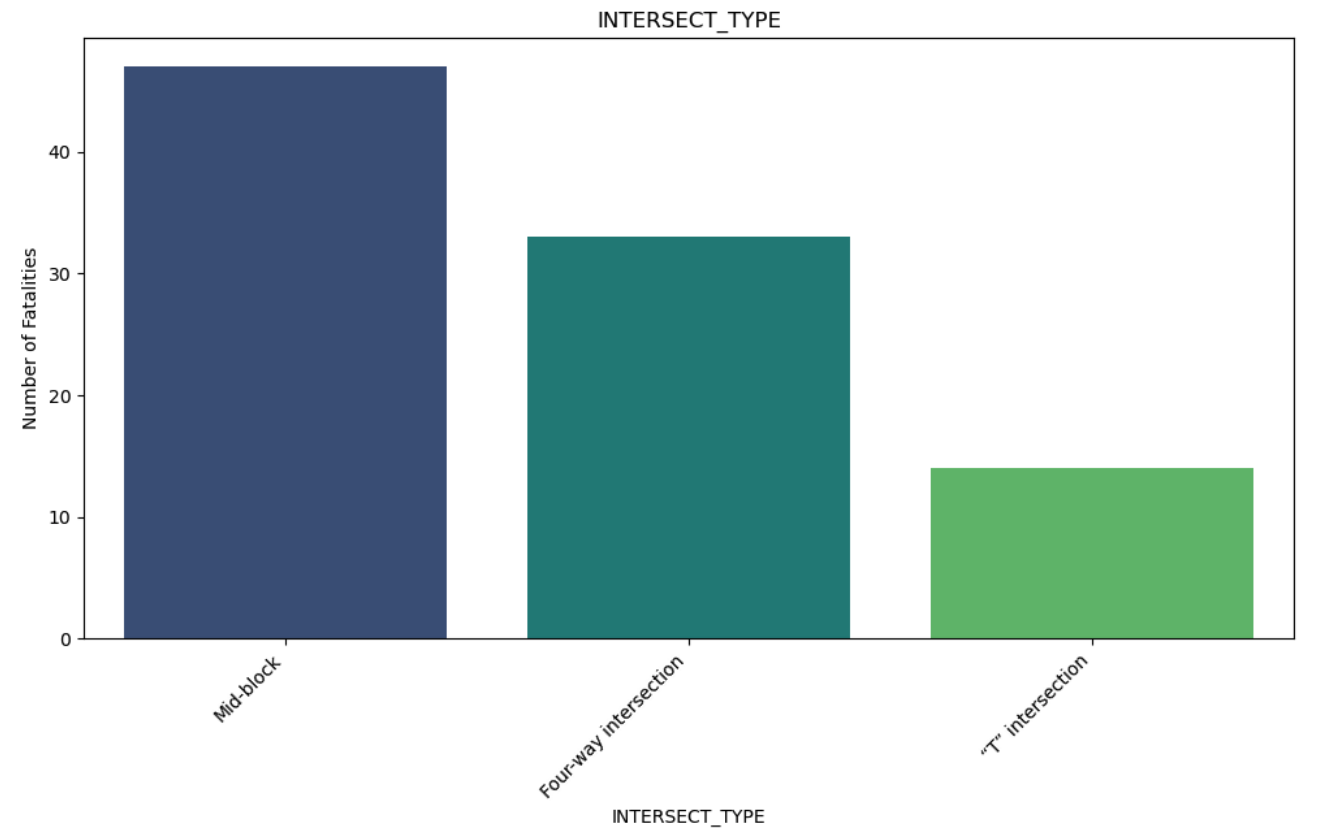
- 94.41% of bicycle fatalities occur in urban settings.
- 5.59% of bicycle fatalities occur in rural settings.

URBAN_RURAL_Dist



INTERSECTION DYNAMICS

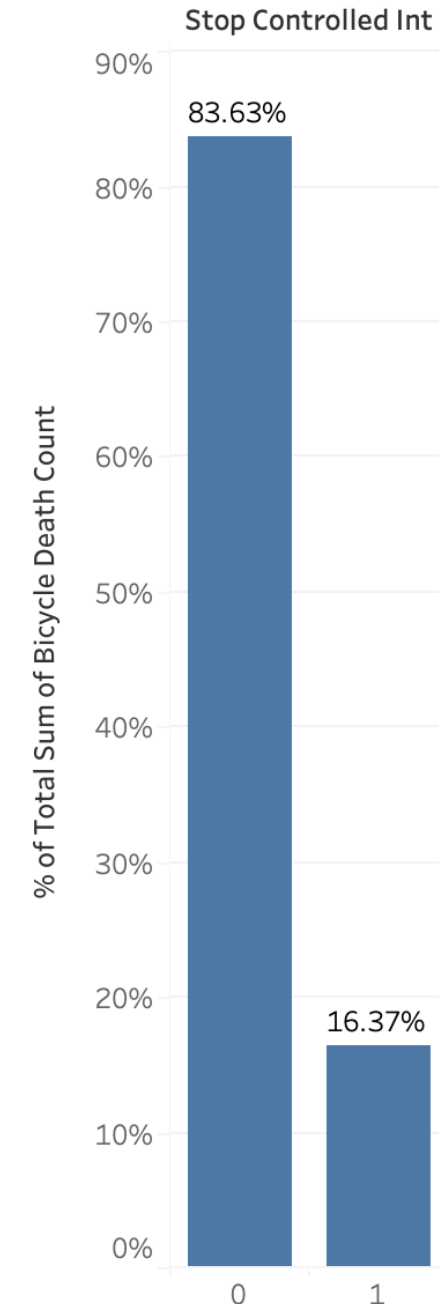
- 51% of bicycle fatalities occur at Mid-Block locations.
- 35% of bicycle fatalities occur at Four-Way Intersections.
- 14% of bicycle fatalities occur at Signalized Intersections.



STOP SIGNS AS LIFESAVERS

- A compelling revelation unfolds as we observe an 84% reduction in the likelihood of severe injuries or death for cyclists at intersections equipped with stop signs.

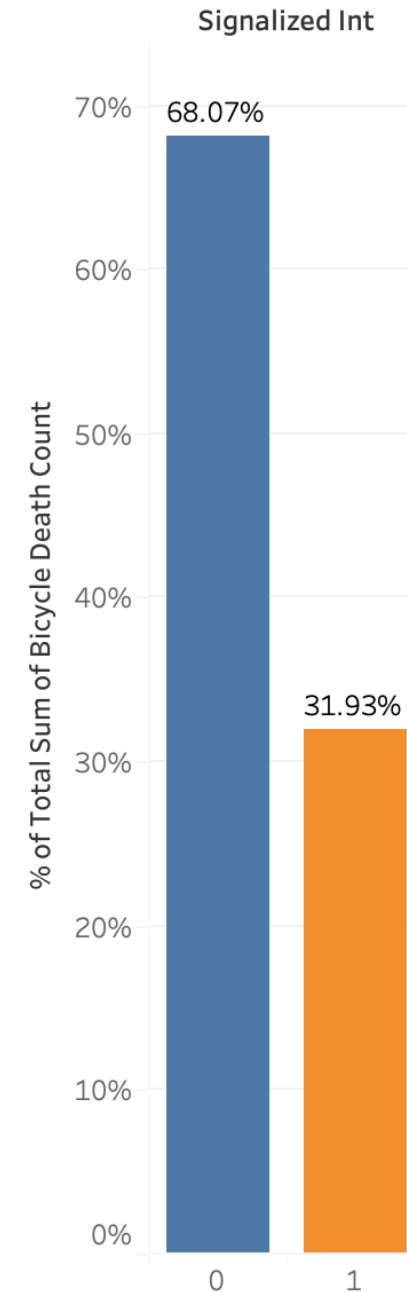
Stop Sign



SIGNALIZED INTERSECTIONS

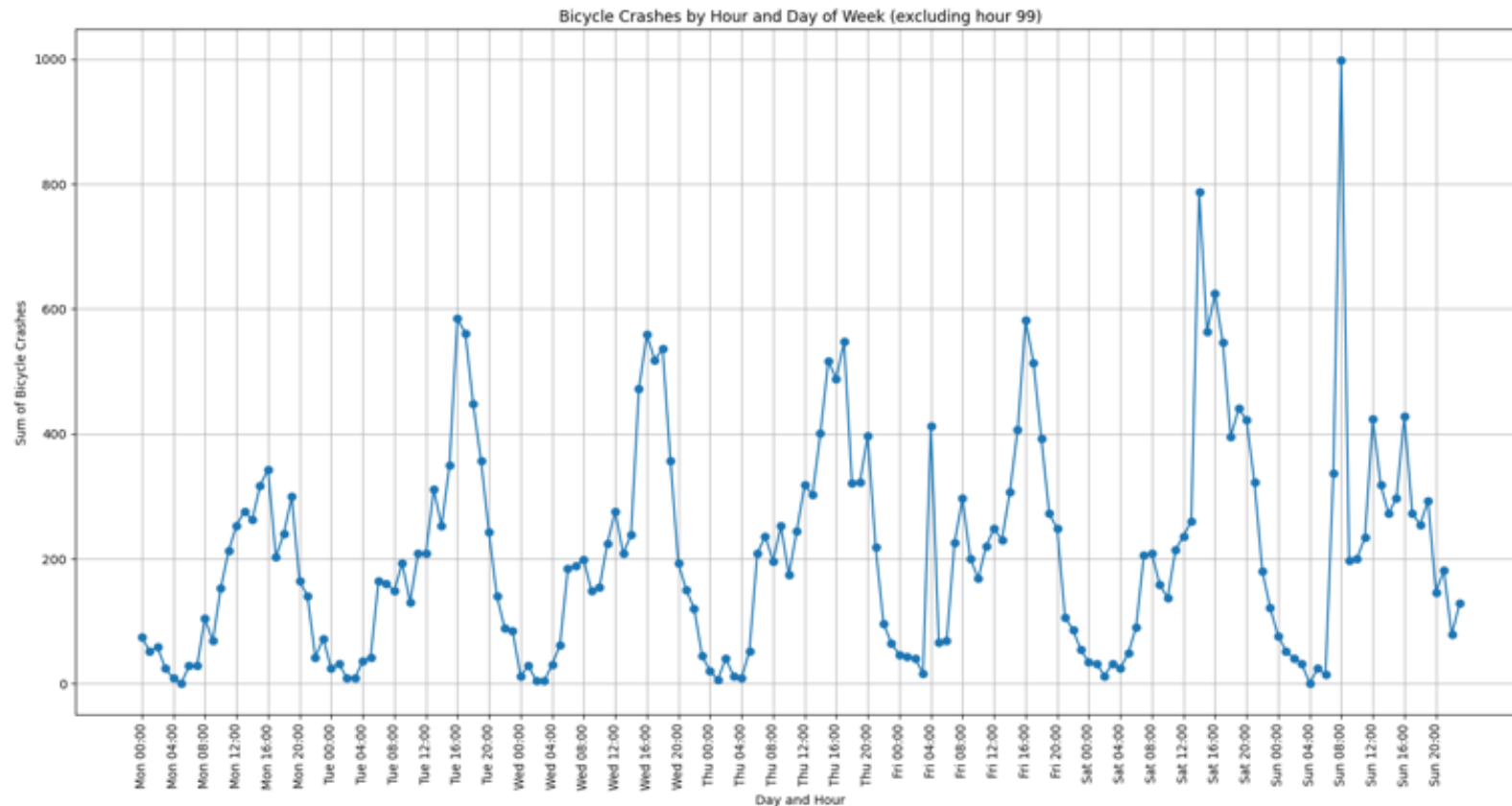
- 30% of bicycle fatalities occur at signalized intersections.
- 70% of bicycle fatalities occur at intersections without signals.

Signalised Intersection



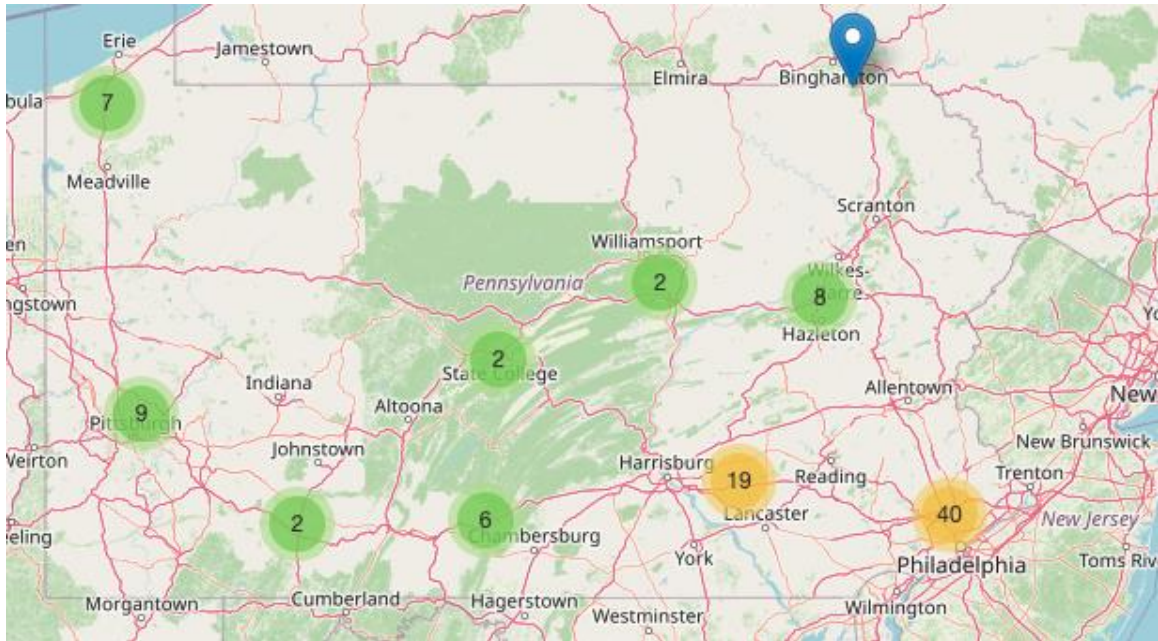
ADDITIONAL INSIGHTS

Speeding, alcohol and drug involvement, mobile phone usage, tailgating, aggressive driving, and illumination, while important facets of road safety, do not emerge as primary catalysts for severe injuries or death in the cyclist demographic.

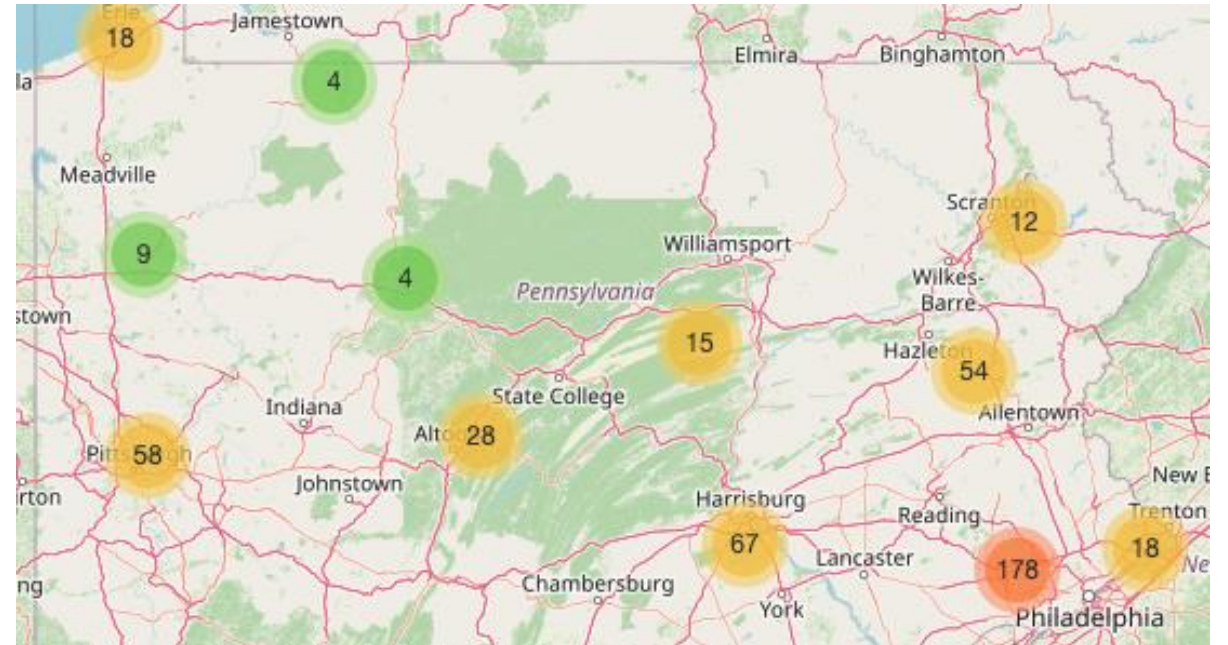


CLUSTER MAP

Fatal incidents



Severe incidents



LOGISTIC REGRESSION

- Two models created
 - Model 1: Balanced Precision and Recall

Logistic Regression:				
	precision	recall	f1-score	support
0	0.89	0.89	0.89	6482
1	0.21	0.21	0.21	923
accuracy			0.80	7405
macro avg	0.55	0.55	0.55	7405
weighted avg	0.80	0.80	0.80	7405

- Model 2: Precision Optimization

Logistic Regression:				
	precision	recall	f1-score	support
0	0.88	1.00	0.93	6482
1	1.00	0.00	0.00	923
accuracy			0.88	7405
macro avg	0.94	0.50	0.47	7405
weighted avg	0.89	0.88	0.82	7405

Odds Ratio:

Feature	Odds Ratio
ILLUMINATION	1.291161
VEH_ROLE	1.099126
SEX_M	1.073329
DISTRICT	1.058815
CURVED_ROAD	1.050806
HIT_PARKED_VEHICLE	1.049742
LOCAL_ROAD	1.048448
LANE_DEPARTURE	1.047643
LEFT_TURN	1.045360
RUNNING_STOP_SIGN	1.038265

Ex: A bicycle-related incident increases by approximately 29.12% for each one-unit increase in the ILLUMINATION feature.



RANDOM FOREST

- Model 1:

	precision	recall	f1-score	support
0	0.89	0.99	0.94	6482
1	0.66	0.17	0.27	923
accuracy			0.89	7405
macro avg	0.78	0.58	0.60	7405
weighted avg	0.86	0.89	0.85	7405

- Feature Importance:

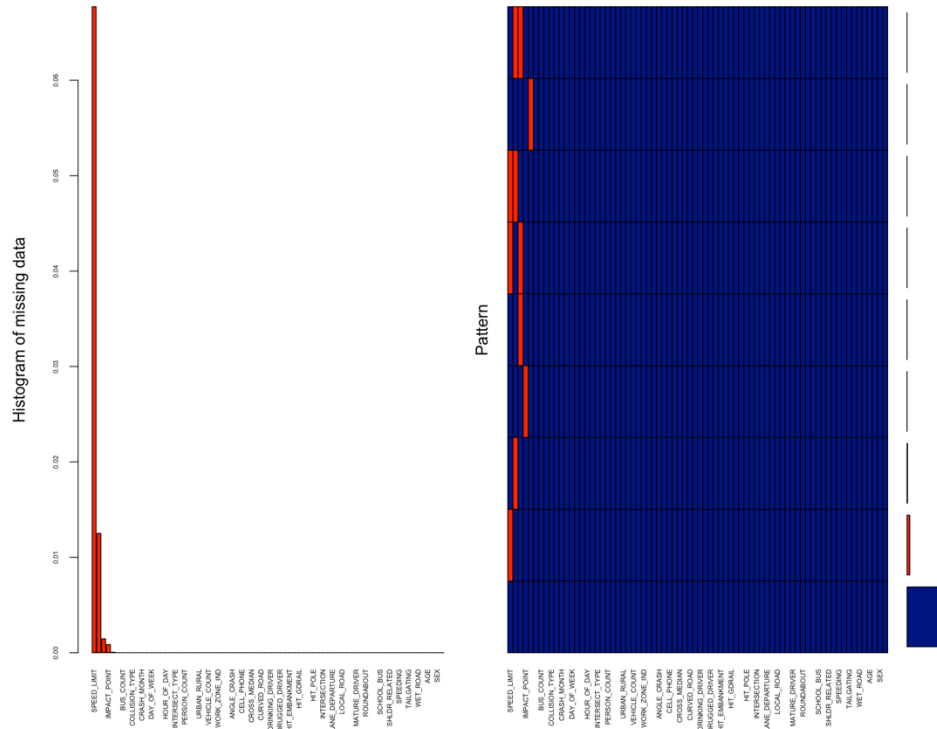
Feature	Importance
URBAN_RURAL	0.108697
COUNTY	0.101325
ILLUMINATION	0.071467
DISTRICT	0.062476
HOUR_OF_DAY	0.055354
COLLISION_TYPE	0.053815
Arrival_Dispatch_TM_Diff	0.049309
LANE_DEPARTURE	0.041296
PERSON_COUNT	0.040961
HIT_PARKED_VEHICLE	0.032741

- Urban and County features have the most impact on the model predictive power.

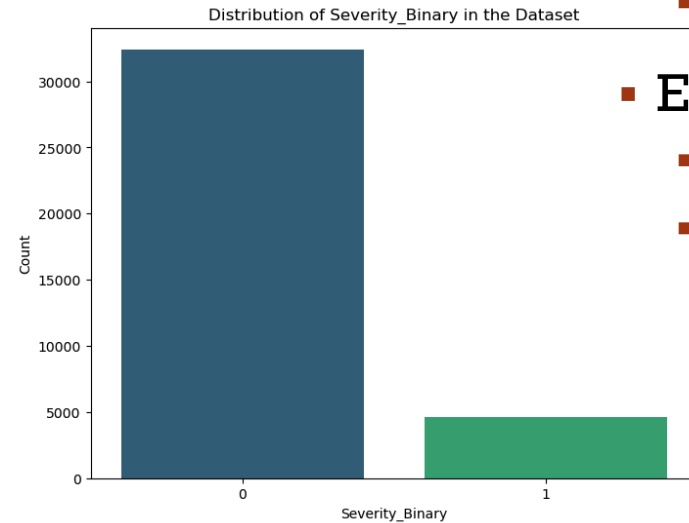


LIMITATIONS

- Data Quality and Completeness



- Imbalanced dataset



- Temporal Scope:

- Data collected from 2018-2022

- External Variables:

- Economy
- Covid



KEY FINDINGS

- Random Forest is the model of choice
 - 89% accuracy with high precision
 - Urban/Rural are the most influential
 - RESTRAINT_HELMET feature was not influential
- 51% of death occurs on Mid-block locations
- 94.41% of severe/fatal incidents occur in Urban settings
- 70% of death occur in absence of signalization
- Majority of severe/fatal incidents occur between 4-6pm.
- Surprisingly speeding, alcohol, aggressive driving, and phone features are not leading contributors for cyclist incidents.

